LNIS – Applicable Document 1 – Volume A – Annex 3 – PRN Spreading Codes README

Note: this document is referenced to as [Annex3] within LNIS Applicable Document 1 – Volume A.

Hexadecimal Codes

The complete set of LunaNet AFS spreading codes for PRNs 1-210 are provided electronically in HEX (hexadecimal format) for development purposes. This allows the implementation of the spreading codes as memory codes. Alternatively, the codes can be generated in real time using the indexed code approach described in LNIS AD1 Volume A.

The names of the files containing the codes are listed in Table 1, and provided electronically as attachments to LNIS AD1 Volume A.

Table 1. List of Spreading Code Electronic Files

Signal	Code	Code Length [symbols]	Filename
AFS-I	Primary Code	*2046*	*#1_GoldCode2046hex210prns.txt*
AFS-Q	Primary Code	*10230*	*#2_l1cp_hex210prns.txt*
	Tertiary Code	*1500*	*#3_Weil1500hex210prns.txt*

Note that codes with length of 2046 and 10230 symbols are not divisible by 4. In order to represent these codes in hexadecimal format and provide them electronically, two zeros are added to the MSB (i.e., to the lefthand side) for all PRNs 1-210. To recover the 2046-chip code for the data channel in binary format, the hexadecimal format can be converted to 2048 binary bits and the two MSBs (leftmost bits) removed. To recover the 10230-chip Q channel code, the hexadecimal format can be converted to 10232 binary bits and the two MSBs removed.

Binary Codes

For completeness, the AFS-Q secondary codes are provided in binary format in Table 2, as well as in LNIS AD1 Volume A.

Table 2. AFS-Q Secondary Codes

Secondary Code Identifier	Secondary Code (binary)
S_0	1110
S_1	0111
S_2	1011
S_3	1101

Each PRN from Table 2 is assigned to each secondary code S_k with k > 3 according with (Eq. 1) below for initial development.

$$k = \begin{cases} i - 1 & \text{for } i \leq 4 \\ \text{mod}(i - 1, 4) & \text{for } i > 4 \end{cases} \tag{1}$$